2. Data, Dataset, Data Warehouse

Everything that concerns the raw data

Types of Datasets

Structured datasets
Organized in rows and columns (like Excel, SQL tables).
Examples: financial records, customer databases.

Time-series datasets
Data collected over time intervals.
Examples: stock prices, weather data, IoT sensors.

Semi-structured datasets
Some organizational properties but not fully tabular.
Examples: JSON, XML, log files, Graph and network.

Unstructured datasets
No fixed schema, free-form.
Examples: images, videos, text documents, audio recordings.

Spatial / Geospatial datasets
Related to location and geography.
Examples: maps, GPS data.

Multimodal datasets
Combine multiple formats.
Example: dataset with images + captions + audio.

Types of Datasets

Structured datasets
Organized in rows and columns (like Excel, SQL tables).
Examples: financial records, customer databases.

Time-series datasets
Data collected over time intervals.
Examples: stock prices, weather data, IoT sensors.

Semi-structured datasets
Some organizational properties but not fully tabular.
Examples: JSON, XML, log files, Graph and network.

Unstructured datasets
No fixed schema, free-form.
Examples: images, videos, text documents, audio recordings.

Spatial / Geospatial datasets
Related to location and geography.
Examples: maps, GPS data.

Multimodal datasets
Combine multiple formats.
Example: dataset with images + captions + audio.

Types of Datasets

Structured datasets
Organized in rows and columns (like Excel, SQL tables).
Examples: financial records, customer databases.

Time-series datasets
Data collected over time intervals.
Examples: stock prices, weather data, IoT sensors.

Semi-structured datasets
Some organizational properties but not fully tabular.
Examples: JSON, XML, log files, Graph and network.

Unstructured datasets
No fixed schema, free-form.
Examples: images, videos, text documents, audio recordings.

Spatial / Geospatial datasets
Related to location and geography.
Examples: maps, GPS data.

Multimodal datasets
Combine multiple formats.
Example: dataset with images + captions + audio.

Key Characteristics of Datasets

Dimensionality

Refers to the number of features (attributes/variables) in a dataset.

Example: An image dataset with 1024 pixels = 1024-dimensional space.
High dimensionality makes data more complex to analyze and visualize.

Curse of Dimensionality

When dimensionality increases:

- Data points become sparse (far apart).
- Distance/similarity measures lose meaning.
- Models require exponentially more data to learn patterns.

Example: A classifier works fine in 2D, but in 100D, you need huge amounts of data.

Key Characteristics of Datasets

Sparsity

Many attributes have zero or missing values.
Sparse datasets are difficult for classical algorithms but well-handled by specialized techniques (e.g., matrix factorization).

Example: In text (bag-of-words, most words absent), recommender systems (most users don't rate most items).

Only Presence Counts

In some datasets, absence of a feature is meaningless.
What matters is which attributes are present, not which are missing.

Example: In a shopping basket dataset, it doesn't matter that you didn't buy all possible products; what matters is what you did buy.

Key Characteristics of Datasets

Resolution

Refers to the granularity or precision of data.
Different resolutions may reveal or hide patterns.

Example: Daily stock prices vs. hourly vs. per second. Average Temperature in 1 Km resolution vs. 5 Km resolution. Patterns vary by resolution

Patterns Depend on the Scale

The scale at which data is analyzed changes interpretation.

Example: Climate trends → at a monthly scale, data shows seasonality; at a yearly scale, long-term warming patterns emerge.

Key Characteristics of Datasets

Distribution

Data values follow certain statistical distributions.
Knowing the distribution is key for selecting algorithms .

Example: Income data is usually skewed, not normal.

Centrality and Dispersion

Centrality: Measures of "center" (mean, median, mode).
Dispersion: Measures of spread (variance, standard deviation, range, IQR).
They describe how data is concentrated or spread out.

Example: Two datasets with the same mean salary but very different variance (one equal, one highly unequal).

Data Objects

Definitions
A datasets is made up of data objects.

A data object represents an entity, the basic unit of a dataset, described by features, and used in analysis, mining, or model training.

A data object (also called a record, instance, example, tuple, sample, or observation) is a collection of attributes (features, variables) values that describe an entity in the dataset.

Object	=	Entity being described	(Database row)
Features	=	Properties of the object	(Database column)

Examples: sales database: customers, store items, sales medical database: patients, treatments university database: students, professors, courses

Data Objects

Example:

Data Object 3 = (C103, Anis L., 22, Algiers, 1200.00)
Features = CustomerID, Name, Age, City, Balance

Customer-ID	Name	Age	City	Balance
C101	Lena H.	19	Algiers	900.75
C102	Yanis B.	31	Oran	850.50
C103	Anis L.	22	Algiers	1200.00

Features

Definitions

Features (or dimensions, attributes, variables): a data field, representing a characteristic or feature of a data object.

Example: customer_ID, name, address, ...

Can be of different types:

- Nominal
- Binary
- Ordinal
- Numeric:
  - Interval-scaled
  - Ratio-scaled

Features

Nominal Feature Type

Categories, states, or names of things

Example: color = {black, red, blue, ...}, marital status = {Married, single, divorced, ...}, occupation = {Engineer, researcher, doctor, ...}, ID numbers, zip codes, ...

Binary Feature Type

Nominal feature with only 2 states (0 and 1)

- Symmetric binary: both outcomes equally important. **Example:** gender
- Asymmetric binary: outcomes not equally important. **Example:** medical test (positive vs. negative) By convention: assign 1 to most important outcome **Example:** HIV positive.

Ordinal Feature Type

Have a meaningful order (ranking) but magnitude between successive values is not known.

Example: Size = {small, medium, large}, satisfaction = {low, medium, high}, army rankings

Features

Numerical Feature Type

To describe a quantity (integer or real-valued).

- **Interval**
  Measured on a scale of equal-sized units: differences are meaningful and comparable everywhere on the scale
  Values have order
  No true zero-point: not a true absence of the quantity

  Example: temperature in C˚or F˚, calendar dates

- **Ratio Inherent**
  We can speak of values as being an order of magnitude larger than the unit of measurement.
  Zero-point: true absence of the quantity

  Example: temperature in Kelvin, length, counts, monetary quantities, age, weight, income,... (10 years is twice as much as 5 years)
  Zero-point: true absence of the quantity

Features

Numerical Feature Type

To describe a quantity (integer or real-valued).

- **Discrete:** Attributes that take on a finite (or countably infinite) number of distinct values.
  Sometimes integer counts or categories.
  Values are separate, no in-betweens.
  Binary attributes are a special case of discrete attributes

  Example: Number of children {0, 1, 2, ... can't be 2.5}. Blood type {A, B, AB, O}.

- **Continuous:** Attributes that take on values from a continuous range, meaning there are infinite possible values within an interval.
  Has real numbers as attribute values

  Example: Height {170.2 cm, 170.25 cm, ...}, Weight (65.00 kg, 65.012 kg, ...), and Temperature (21.3 °C, 21.33 °C, ...).

Similarity and Dissimilarity Measures

Similarity: A measure of how alike two data objects are.

- High similarity = objects are close in terms of their features.
- Value usually lies in the range [0, 1] (1 = identical, 0 = completely different).

Examples:
Two texts with many shared words → high similarity.
Two images with nearly identical pixels → high similarity.

Dissimilarity / Distance: A measure of how different two data objects are.

- High dissimilarity = objects are far apart in terms of attributes.
- Value is often non-negative, sometimes normalized to [0, 1].

Examples:
Euclidean distance between two points in space.
Hamming distance (number of mismatches) between two strings.

Proximity: refers to a similarity or dissimilarity

Similarity and Dissimilarity Measures

Proximity Measure for Binary Features

- A contingency table for binary data
- Distance measure for symmetric binary variables
- Distance measure for asymmetric binary variables

TODO

Similarity and Dissimilarity Measures

Distance (Dissimilarity) for Nominal Features

- **Simple Matching**
  Based on how many attributes match vs. don't match.

$$\textbf{Distance(i,j)} = \frac{\textbf{\# of features} - \textbf{\# of matches}}{\textbf{\# of features}}$$

Example:
Object i = (Red, Cat, Male)
Object j = (Blue, Cat, Male)
# of features = 3
# of matches = 2 (Cat & Male)
Distance = (3-2)/3 = 1/3 = 0.33

Similarity and Dissimilarity Measures

Distance (Dissimilarity) for Nominal Features

- **Hamming Distance**
  Counts the number of positions where attributes differ.

  Distance(i,j) = # of different values

  Example:
  i = (1,0,1,1,0),
  j = (1,1,0,1,0)
  Comparison: (equal, diff, diff, equal, equal)
  Distance = # of diff = 2

- Using a large number of binary attributes
  Creating a new binary attribute for each of the M nominal states
  Use hamming distance or binary feature measures.

Similarity and Dissimilarity Measures

Distance (Dissimilarity) for Numerical Features

- **Minkowski distance:** distance of order p between two objects.

$$d(i,j) = \sqrt[h]{|x_{i1} - x_{j1}|^h + |x_{i2} - x_{j2}|^h + \cdots + |x_{ip} - x_{jp}|^h}$$

where $i = (x_{i1}, x_{i2}, ..., x_{ip})$ and $j = (x_{j1}, x_{j2}, ..., x_{jp})$ are two p-dimensional data objects, and h is the order (the distance so defined is also called L-h norm)

- **Manhattan distance:** city block, L1 norm.

$$d(i,j) = |x_{i_1} - x_{j_1}| + |x_{i_2} - x_{j_2}| + ... + |x_{i_p} - x_{j_p}|$$

- **Euclidean distance:** L2 norm.

$$d(i,j) = \sqrt{(|x_{i_1} - x_{j_1}|^2 + |x_{i_2} - x_{j_2}|^2 + ... + |x_{i_p} - x_{j_p}|^2)}$$

Similarity and Dissimilarity Measures

Distance (Dissimilarity) for Ordinal Features

- **Classical Approach:**
  - Convert categories to ranks / Data encoding
  - Normalize them (0–1 scale),
  - Apply numeric distance (Manhattan, Euclidean, etc.).

- Other specialized measures
  - Spearman,
  - Kendall,
  - Goodman-Kruskal
  - Treat like interval-scaled

  exist for more advanced similarity.

Similarity and Dissimilarity Measures

Distance (Dissimilarity) for Mixed Features

- **Gower's Distance:** similarity S_{ij} between two objects is the weighted average of the similarities calculated for all their features.

$$S_{ij} = \frac{\sum_{k=1}^{p} w_{ijk} s_{ijk}}{\sum_{k=1}^{p} w_{ijk}},$$

where i and j are two data objects with p features, the w_{ijk} are non-negative weights usually set to 1 and s_{ijk} is the similarity between the two objects regarding their k^{-th} feature.

- Other alternatives exists
  Example: HEOM, HVDM, Sum of separately calculated distances.

Dataset Vs Data Warehouse

Dataset

A dataset is a collection of data, usually organized in a table.

Size/Scope: Often small or medium, used for a specific analysis or experiment.

Format: Can be structured (tables, CSV, Excel), semi-structured (JSON, XML), or unstructured (text, images).

Purpose: Directly used for analysis, training models, or testing hypotheses of a specific domain.

Example: A CSV file of 10,000 customer transactions for a given month and store.

Data Warehouse

A large, centralized repository that integrates data from multiple sources / datasets.

Size/Scope: Very large, often terabytes or petabytes; designed to store historical and current data.

Format: Structured, optimized for querying and reporting (uses schemas, OLAP cubes, star/snowflake schema).

Purpose: Supports decision-making, business intelligence, and large-scale data mining by providing a “single unified source of truth”.

Example: A company’s data warehouse storing 5 years of sales, customer, and inventory data from different branches and platforms.

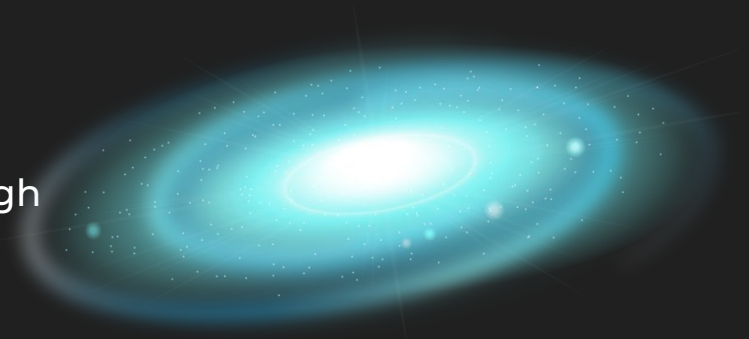
Data Warehouse

	Fact tables: store numeric measurable data (facts) and links it to related dimensions through foreign keys.(quantities, revenue).	**Dimension tables:** store descriptive data that describe the objects involved in the facts. (product details, time periods).
Content	Numeric or quantitative data (measures).	Descriptive, textual, or categorical data.
Purpose	Represents *what happened*.	Provides *context* to the facts (“who, what, where, when, how”).
Grain	Each row corresponds to a specific event (e.g., one sale transaction).	Contains a **primary key** that links to fact tables.
Keys	Contains **foreign keys** referencing dimension tables.	Smaller, fewer rows but more columns (rich descriptions).
Size	Usually the **largest** table (many rows).	Descriptive, textual, or categorical data.

Dataset Vs Data Warehouse

Star Schema: A Simple and Popular Data Warehouse Schema

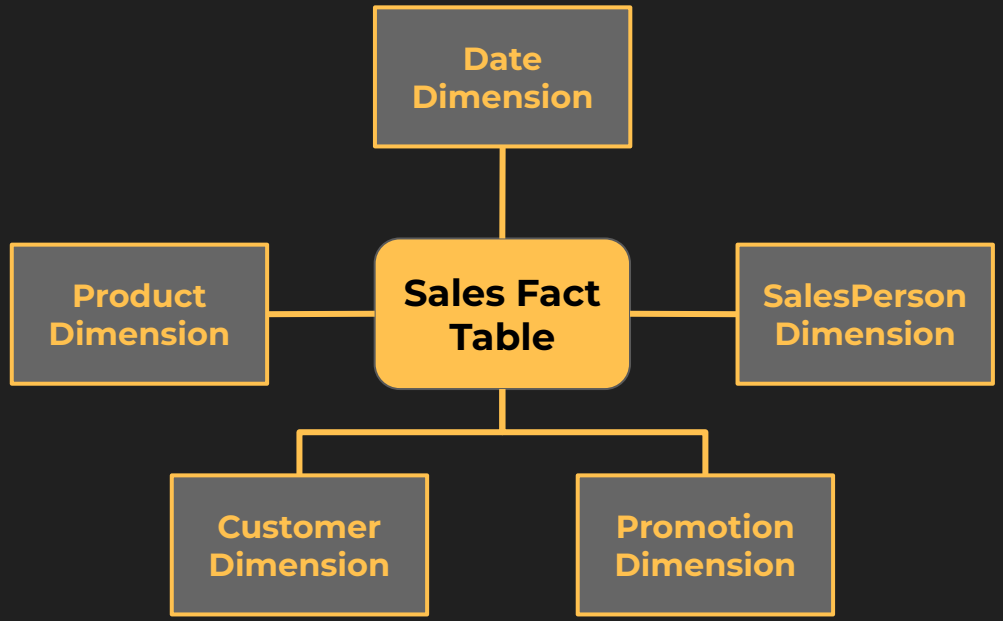
Snowflake Schema: More Organized, But More Work



Galaxy Schema: When One Star Isn't Enough

Dataset Vs Data Warehouse

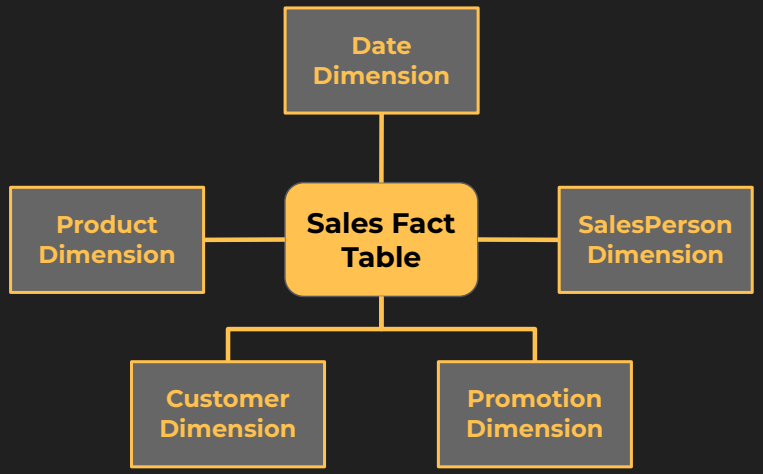
Star Schema: A Simple and Popular Data Warehouse Schema



The fact table holds the key numeric data, while dimensions provide context, making it easy to query and analyze, which is why it is also the most popular schema.

Dataset Vs Data Warehouse

Star Schema: A Simple and Popular Data Warehouse Schema



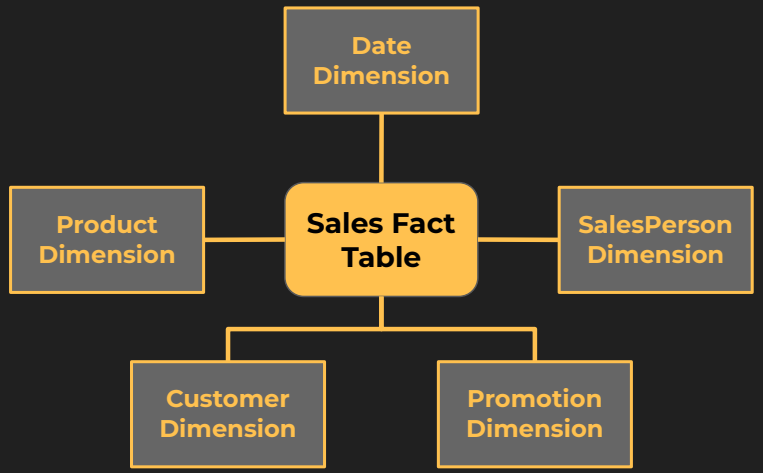
FactSales			
sale_id	date_id	prod_id	…
1	202401	1001	…
2	202401	1002	…
3	202402	1001	…

DimProduct		
prod_id	prod_name	category
1001	Widget A	Widgets
1002	Widget B	Widgets

The fact table holds the key numeric data, while dimensions provide context, making it easy to query and analyze, which is why it is also the most popular schema.

Dataset Vs Data Warehouse

Star Schema: A Simple and Popular Data Warehouse Schema



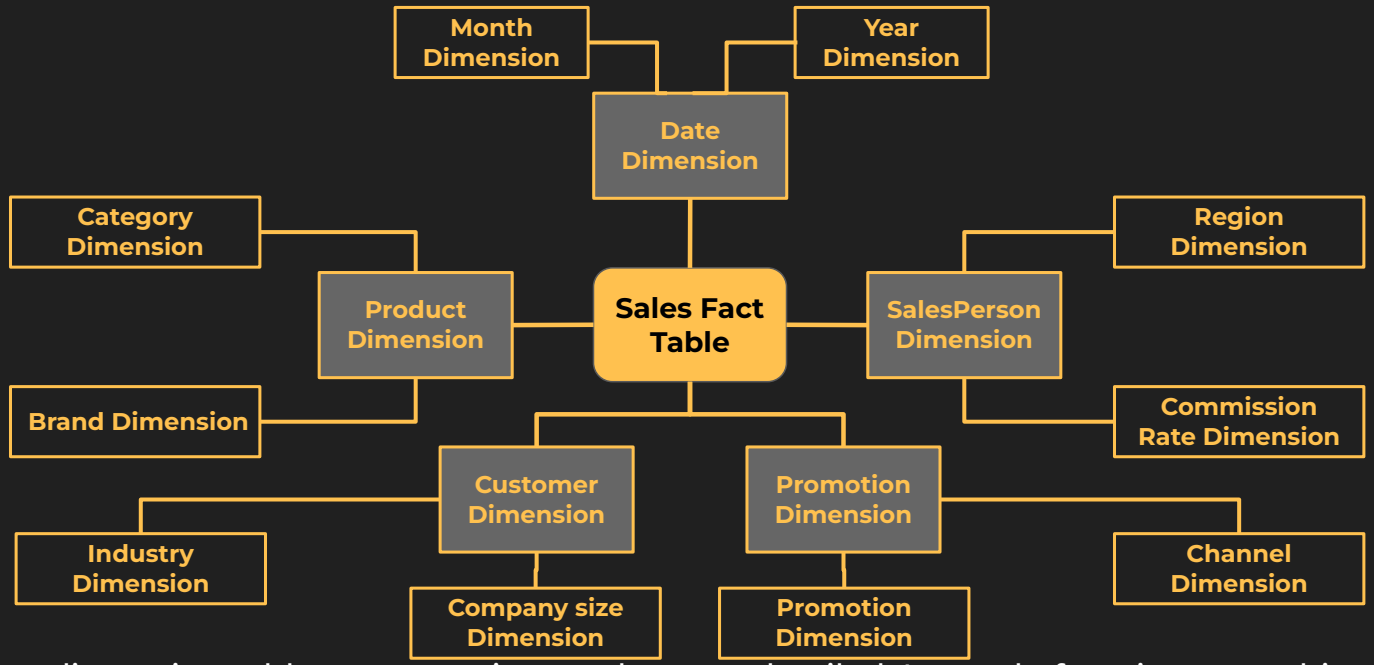
FactSales			
sale_id	date_id	prod_id	…
1	202401	1001	…
2	202401	1002	…
3	202402	1001	…

DimProduct		
prod_id	prod_name	category
1001	Widget A	Widgets
1002	Widget B	Widgets

```
SELECT p.prod_name, COUNT(s.sale_id) as total_sales
FROM FactSales s
JOIN DimProduct p ON s.prod_id = p.prod_id
GROUP BY p.prod_name;
```

Dataset Vs Data Warehouse

Snowflake Schema: More Organized, But More Work



When dimension tables start getting too large or detailed. Instead of storing everything in a single dimension table, you “normalize” it across multiple, smaller related tables.

Dataset Vs Data Warehouse

Snowflake Schema: More Organized, But More Work

DimBrand		
brand_id	**brand_name**	**product_type**
200	SimpleShoes	Apparel
201	FutureFeet	Electronics

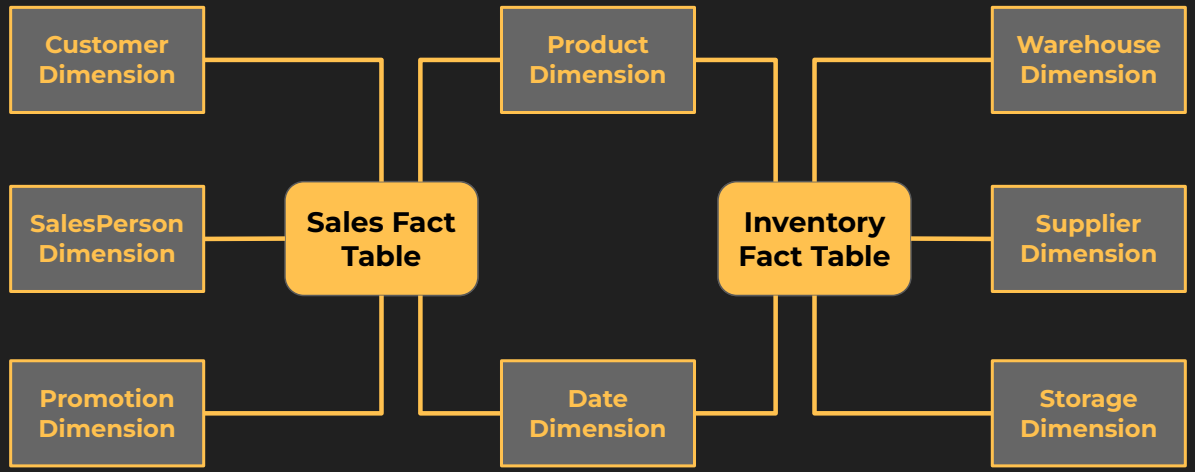
FactSales			
sale_id	**date_id**	**prod_id**	**…**
1	202401	1001	…
2	202401	1002	…
3	202402	1001	…

DimProduct		
prod_id	**prod_name**	**brand_id**
1001	Super Sneakers	200
1002	Clunky Clogs	200

```
SELECT b.brand_name, p.prod_name, COUNT(s.sale_id) AS total_sales
FROM FactSales s
JOIN DimProduct p ON s.prod_id = p.prod_id
JOIN DimBrand b ON p.brand_id = b.brand_id
GROUP BY b.brand_name, p.prod_name;
```

Dataset Vs Data Warehouse

Galaxy Schema: When One Star Isn't Enough



Star Schema with multiple fact tables that share dimensions. That way you can unify some data sources while still maintaining separate facts.

Dataset Vs Data Warehouse

Online analytical processing (OLAP) Cube

- Structure
- Operations
- Example (3D)

TODO

ETL Vs ELT Pipeline

Extract, Transform, Load (ETL)

ETL is the pipeline that moves raw data from many sources, cleans it up, and delivers it into a warehouse so analysts and data mining tools can use it.

Extract, Load, Transform (ELT)

ELT is the pipeline that moves raw data from many sources into the data warehouse first, and then cleans and transforms it inside the warehouse, depending on the task needed by analysts.

ETL Vs ELT Pipeline

Extract, Transform, Load (ETL)

Extract:

- Data is collected from multiple source systems (databases, files, APIs, logs, CRM, sensors...).
- Different sources may have different formats, encodings, or data quality issues.
- Goal: Gather raw data without losing anything important.

Transform:

- Data cleaning (handling missing values, duplicates, errors).
- Data integration (combining different sources).
- Data conversion (changing formats, units, or codes).
- Aggregation (e.g., summing daily sales into monthly totals).

Load:

- Full load: All data is loaded at once (first time).
- Incremental load: Only new/changed data is added (daily, hourly, real-time).

ETL Vs ELT Pipeline

Aspect	ETL	ELT
Transformations	Before loading	After loading
Best for	On-premise warehouses (limited storage/compute)	Cloud warehouses (scalable, high compute power)
Performance	May be slower if data is huge (transform step before load)	Faster for large data, since warehouse does transformations
Data Storage	Only transformed data stored	Raw + transformed data stored (flexibility)
Flexibility	Fixed transformations, harder to reprocess	Can re-transform anytime (raw data is available)